ARPIT DAVE

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PROFESSIONAL SUMMARY:

- Around 10+ years of **IT experience** as a **Data Scientist in AI/ML/GenAI**, including profound expertise and experience on statistical data analysis such as transforming business requirements into analytical models, designing algorithms, and strategic that scales across massive volumes of data.
- Hands-on experience building autonomous AI agent workflows using LangGraph, CrewAI, AutoGen, and LlamaIndex Agents, orchestrating multi-step reasoning, tool use, and human-in-the-loop patterns for complex business processe
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines including advanced Graph RAG architectures with Neo4j and knowledge graphs to deliver context-rich, grounded responses from enterprise data.
- Experienced with agentic design patterns including ReAct, Plan-and-Execute, reflection loops, and multi-agent collaboration for end-to-end task automation and intelligent orchestration.
- Experienced in designing end-to-end AI/ML solutions on Microsoft Azure, leveraging Azure OpenAI Service, Azure AI Search (formerly Cognitive Search), Azure Machine Learning, Azure Databricks, and Azure AI Studio for building, deploying, and scaling enterprise-grade AI applications.
- Skilled in Azure DevOps and CI/CD pipelines for MLOps, enabling automated model training, versioning, deployment, and monitoring using Azure ML Pipelines, MLflow, and Azure Kubernetes Service (AKS)
- Proficient in vector databases (Pinecone, Weaviate, ChromaDB, FAISS) and embedding strategies for semantic search, document retrieval, and long-term agent memory.
- Proficient in Statistical Methods like Regression models, hypothesis testing, confidence intervals, principal component analysis and dimensionality reduction.
- Professional working experience in Machine Learning algorithms such as, linear regression, logistic regression, Naive Bayes, Random Forests, Decision Trees, Clustering and Principal Component Analysis.
- Strong background in evaluating machine learning models using various metrics and methodologies to ensure robust and reliable predictive capabilities.
- Experienced with advanced NLP methods including sentiment analysis, text classification, named entity recognition, and fine-tuning LLMs (GPT-4, Claude, Llama, Mistral) for domain-specific tasks.
- Adept in using Hadoop Distributed File System (HDFS) for scalable and fault-tolerant data storage, and leveraging Map Reduce for distributed data processing across large datasets.

SKILL SET

Generative AI, LLMs (GPT-4o, Claude, Llama, Mistral), Agentic AI, LangChain, LangGraph, CrewAI, AutoGen, Semantic Kernel, RAG, Graph RAG, Prompt Engineering, Machine Learning, Deep Learning, Statistics, Python, R, Spark, SQL, Natural Language Processing (NLP), Hugging Face Transformers, BERT, SpaCy, NLTK, Word Embeddings, Sentiment Analysis, Text Mining, TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Pandas, Dask, Neo4j, Pinecone, Weaviate, ChromaDB, FAISS, Snowflake, Delta Lake, Microsoft Azure (OpenAI Service, AI Search, AI Foundry, Databricks, AKS, Data Factory, Synapse), Docker, Kubernetes, Azure DevOps, GitLab CI/CD, MLflow, LangSmith, Flask, FastAPI, REST APIs, Postman, Tableau, Power BI, Matplotlib, Seaborn, Plotly, Time Series Forecasting, ARIMA, SARIMA, Predictive Models, Exploratory Data Analysis (EDA), Supervised and Unsupervised Learning, A/B Testing, Financial Analytics, Statistical Analysis.

PROFESSIONAL EXPERIENCE:

Squad Soft Inc **2023 - Present**
Role: Sr. Data Scientist
Responsibilities:

- **Designed and built agentic AI architectures for enterprise-scale** knowledge base and graph construction, powering an AI description generator followed by automated classification pipelines — reducing manual data classification effort by 70%, **improving classification accuracy by 35%, and cutting time-to-insight from weeks to hours.**
- **Developed efficient AI-driven classification leveraging similarity scores,** Maximal Marginal Relevance (MMR), and cosine similarity — achieving **94% classification accuracy,** reducing false positives by 40%, and scaling retrieval precision across 5M+ records with sub-second response times.
- **Created an agentic tool for automated bug detection**, architecture design, and blueprint generation on Azure DevOps — reducing bug triage time by 55%, accelerating architecture documentation by 3x, and eliminating 80% of manual ticket creation overhead.
- **Architected the AI-Driven Data Classification Pipeline** (raw data ingestion → AI description generation → knowledge base augmentation → SME approval → CID classification via RAG → KD element mapping → RBAC enforcement) — processing 10M+ records daily, reducing SME review cycles by 60%, and achieving 92% first-pass classification accuracy.
- **Built the AI Epic Management Panel** — a GitLab-integrated agentic workspace with multi-agent orchestration (Orchestrator, Description Agent, Classification Agent, KD Mapping Agent), self-critique loops, one-click publishing, and blueprint diagramming via LangGraph and Semantic Kernel — cutting epic creation time from 4 hours to 15 minutes, improving documentation consistency by 85%, and enabling 50+ engineers to self-serve.
- Engineered multi-agent systems using LangGraph, CrewAI, and AutoGen with tool-calling capabilities, memory persistence, and reflection patterns — automating 12+ complex enterprise workflows, reducing human intervention by 75%, and saving approximately 2,000+ engineering hours annually.

- Developed a high-accuracy Question Answering system leveraging LangChain, Azure OpenAI (GPT-4o), and Graph RAG with Neo4j — achieving ~90% extraction accuracy across 50K+ documents, reducing research time by 65%, and supporting 500+ concurrent users with <2s latency.
- Built production-grade RAG pipelines using Azure AI Search (hybrid + vector search), Azure OpenAI embeddings, and reranking strategies (Cohere Rerank, cross-encoders) — i**mproving retrieval relevance by 45%, reducing hallucination rates by 60%, and handling 1M+ document chunks with 99.5% uptime.**
- Developed advanced techniques for reducing the computational load of LLMs — including quantization (GPTQ, AWQ), **LoRA/QLoRA fine-tuning,** and speculative decoding — reducing inference costs by 50%, cutting latency by 40%, and maintaining 98% of baseline model performance.
- Led the design and implementation of a Multichannel/Omnichannel Decision Engine, integrating machine learning models with real-time event streaming (Kafka, Azure Event Hubs) — processing 2M+ events daily, improving decision throughput by 4x, and reducing rule-execution latency from minutes to milliseconds.
- Implemented advanced statistical models such as Adstock and Hill functions within Marketing Mix Modeling (MMM) frameworks **— improving media attribution accuracy by 30%, optimizing $15M+ in annual ad spend allocation, and reducing wasted media budget by 22%.**
- Led development of a BYOML (Bring Your Own ML) platform on Azure ML and Databricks — onboarding 100+ data scientists, reducing **model deployment time by 60% (from 3 weeks to 5 days), and enabling 200+ custom models in production with standardized CI/CD pipelines.**
- Designed and deployed model monitoring and observability **using LangSmith, Azure Monitor, and custom drift detection pipelines — catching 95% of data drift incidents within 1 hour, reducing model degradation incidents by 70%**, and maintaining SLA compliance of 99.9% across all production models.
- Architected scalable AI solutions on Azure Cloud Services, leveraging Azure OpenAI Service, Azure AI Foundry, Azure Kubernetes Service (AKS), and **Azure Cosmos DB — supporting 10,000+ API requests per minute, achieving** 99.95% availability, and reducing infrastructure costs by 35% through auto-scaling and spot instances.
- Built event-driven data pipelines using Azure Data Factory, Azure Synapse Analytics, and Delta Lake on Databricks — processing 5TB+ daily, reducing ETL pipeline runtime by 55%, and enabling near real-time data freshness with <10-minute end-to-end latency.
- Managed SQL databases with MySQL and PostgreSQL, optimizing query performance through indexing strategies, partitioning, and materialized views — **improving query execution time by 60%, reducing database costs by 25%, and ensuring zero-downtime migrations across distributed systems.**

Accenture **Jun 2021 – June 2022**
Role: Data Scientist
Responsibilities:

- **Led the design and development of an intelligent budget planning application** for a global biopharmaceutical company, **standardizing processes across 20+ markets** and integrating data from multiple disparate systems.
- **Implemented advanced multivariate regression and time series models** to provide accurate budget forecasts, significantly enhancing financial planning and decision-making.
- **Engineered a scalable back-end using Python (AWS Lambda) with RESTful APIs**, ensuring efficient data processing and seamless integration with various data sources.
- **Developed a dynamic front-end with AngularJS**, delivering an intuitive and responsive user interface for an enhanced user experience.
- **Leveraged AWS services** such as **S3 (using Boto3)** for secure data storage and retrieval, **CloudWatch for log monitoring**, and **MySQL for robust database management**, building a resilient cloud-based solution.
- **Utilized the Domino Platform for job scheduling and orchestration**, automating workflows and improving operational efficiency across the application lifecycle.
- Designed and deployed a set of **Python** modelling **APIs** for customer analytics, which integrate multiple machine learning techniques for various user behavior prediction.
- Incorporated various machine learning algorithms and advanced statistical analysis like **decision trees, regression models, SVM, clustering** using **Scikit-learn** package in **Python.**
- Performed technical build out, created **Tableau dashboards** to enable reports on data analysis and modeling. Implemented data acquisition and manipulation based on **SQL.**
- Implemented **RNNs** for sequential data analysis, including applications in time series forecasting and natural language processing.
- Created dashboards with **Matplotlib** and **Seaborne**, to communicate insights and trends to stakeholders effectively.
- Collaborated with data engineering teams to integrate **Databricks** workflows with **Azure Data Factory** and **Azure SQL Database** for seamless data movement and processing.
- Developed and maintained reusable TensorFlow components and libraries to streamline the model development process.
- Created and executed training workflows, including data preprocessing, model training, evaluation, and deployment using TensorFlow Extended (TFX).
- Utilized **Tableau Prep** to clean, shape, and merge data sources for more effective analysis and visualization, ensuring high data quality and consistency across all reporting platforms.
- Implemented **natural language processing (NLP)** solutions using **NLTK** and **SpaCy** for text analysis, sentiment analysis, and language modeling.

Cognizant Technology Solution **Jul 2020 – May 2021**
Role: Lead Data Scientist

- **Responsibilities:** Used **Power BI's** custom visuals marketplace to create unique and tailored visual representations of data.
- Developed interactive **storytelling techniques** to present complex data insights in a clear and engaging manner, aiding decision-making processes.
- Utilized **AWS Lambda** for server less data processing tasks, integrating with **AWS** services such as **AWS S3** and **AWS Dynamo DB** for real-time analytics.
- **Developed machine learning models to predict diabetes risk**, enabling early detection without the need for a doctor's visit, which is especially crucial during healthcare crises like the **COVID-19 pandemic**.
- **Utilized a range of algorithms** including **Logistic Regression**, **k-Nearest Neighbors (k-NN) with Jaccard distance**, **Decision Trees**, **Random Forests**, and **Ensemble Models** to optimize predictive accuracy and sensitivity for binary classification.
- **Conducted extensive data handling and exploratory data analysis** on features such as **age**, **gender**, and clinical symptoms (**polyuria**, **polydipsia**, **polyphagia**, **visual blurring**), identifying key predictors and correlations associated with diabetes.
- **Discovered critical insights** like the strong correlation of **polydipsia** and **polyuria** with diabetes risk, while finding that **obesity** was not a significant predictor, contrary to common expectations in the medical community.
- **Optimized models using cross-validation and hyperparameter tuning**, achieving high performance metrics with **Logistic Regression (95.5% accuracy, 97.6% sensitivity)** and further enhancing predictive power through **Ensemble Models**.
- **Visualized data and model outcomes** using **correlation heatmaps**, **ROC curves**, **AUC**, **precision**, **recall**, and **F1-scores**, ensuring both the interpretability and reliability of the models for potential clinical application. **Demonstrated the utility of machine learning in bridging healthcare gaps**, with plans to **incorporate more diverse datasets** and **develop a user-friendly application** for widespread, accessible diabetes risk assessment, ultimately aiming to improve patient outcomes and reduce undiagnosed cases.

Cognizant **Apr 2018 – Jun 2020**
Role: Data Scientist
Responsibilities:

- **Led a team to develop an intelligent Health Care Incentive Compensation Application**, utilizing **IQVIA sales data** demonstrating a **deep understanding of the life sciences industry**.
- **Achieved 100% accuracy and reduced turnaround time by 630%** through **process standardization and automation**, significantly improving operational efficiency.
- **Engineered robust ETL pipelines using Informatica Intelligent Cloud Services (IICS) and Snowflake**, processing and harmonizing large volumes of raw IQVIA data for efficient storage and retrieval.
- **Built interactive dashboards and reports in Power BI**, implementing **row-level security** and providing stakeholders with secure, actionable insights.
- Designed and implemented end-to-end systems for Data Analytics and Automation, integrating custom visualization tools using **R** and **Power BI.**
- **Developed a predictive model to generate a "Patient Non-Adherence Risk Score"**, utilizing patient demographics, treatment specifics, and historical adherence data to identify patients at high risk of non-adherence.
- **Addressed the strategic problem of poor drug adherence**, which leads to adverse health outcomes and significant revenue loss—45% of patients discontinue therapy within 90 days.
- **Collaborated with a Specialty Pharmacy client** to implement a machine learning framework that continuously scores patients and tailors interventions, measuring effectiveness through metrics like patient Days of Therapy (DOT).
- **Integrated diverse data sources** including medical and prescription claims, patient activity statuses, physician data, Medicare Part D, and health plans to enrich the model's predictive capabilities.
- **Applied features engineering, refined reature from 162 to 18 key predictors**, identifying significant drivers of non-adherence such as nurse interactions, therapy adjustments, and health plan changes, using techniques like Kaplan-Meier survival analysis.
- **Implemented supervised machine learning models**—Logistic Regression, Random Forest, Neural Networks—to predict non-adherence, achieving high model performance (AUC) and classification accuracy.
- **Demonstrated measurable impact** by increasing incremental Days of Therapy and revenue recovery for the client, enhancing patient outcomes through targeted interventions based on the risk scores.
- Optimized machine **learning** models through **hyper parameter tuning** and **grid search techniques**. Applied **regression models** to predict continuous outcomes and evaluate the relationship between variables.
- Applied classification algorithms to categorize data into **predefined classes, including decision trees, random forests, support vector machines** and **k-nearest neighbors.**
- Managed **SQL** databases with **MySQL** and **PostgreSQL**, optimizing query performance and ensuring data integrity.
- Used **Power BI's** custom visuals marketplace to create unique and tailored visual representations of data.
- Developed interactive **storytelling techniques** to present complex data insights in a clear and engaging manner, aiding decision-making processes.
- Utilized **AWS Lambda** for server less data processing tasks, integrating with **AWS** services such as **AWS S3** and **AWS Dynamo DB** for real-time analytics.
- Applied advanced data manipulation and analysis techniques using **Pandas** and **Dask**, enhancing the efficiency of data processing and analysis pipelines.

Cognizant Technology Solution **Apr 2016 – Mar 2018**
Role: Data Science Analyst
Responsibilities:

- **Developed machine learning models to predict patient adherence and optimize marketing strategies**, significantly enhancing prescription rates and improving patient health outcomes in the **life sciences sector**.

- **Implemented advanced algorithms** like **Logistic Regression**, **Random Forests**, and **Gradient Boosting Machines**, achieving high model performance metrics (**AUC**, **accuracy**) in predicting patient adherence and assessing marketing effectiveness.
- **Formulated and tested key hypotheses**:
- **Payor Impact Hypothesis**: Demonstrated that **co-pay card usage significantly increases patient adherence**, highlighting the importance of affordability in treatment continuity.
- **Marketing Strategy Hypothesis**: Evaluated the effectiveness of various **marketing strategies** on prescription rates using metrics from **online engagements**, **traditional events**, and **outreach efforts**.
- **Optimized resource allocation by leveraging segmentation and model insights**, focusing on high-impact healthcare professionals (HCPs) and patient groups, which resulted in:
- **Improved prescription rates** and enhanced patient adherence.
- **Optimize marketing spend** and better **ROI** for marketing campaigns.
- Performed data analysis and data profiling using complex **SQL** queries on various sources systems including **Oracle** and **SQL Server,** also writing the **stored procedure for the end-to-end automation for sales crediting, goal calculation and attainment.**
- Automated data extraction and transformation processes through **Python** scripts to streamline data workflow

AWARDS & RECOGNITION

- **Recognized by Delivery Excellence Award - 2020 (For outstanding performance in deploying innovative data solutions.)**
- **Recognized by CDB Applause Rewards and recognition Shining Star Award - 2019 (Recognized for leading best practices in data Analytics)**
- **Recognized by CDB Applause Rewards and recognition - Xtramile Award winner-2018**
- **Recognized by Outreach -Wheel of Wisdom 2018.**

EDUCATION

Case Western Reserve University, School of Engineering, Cleveland, OH
Master of Science in Engineering: Computer Science GPA: 3.50/4.00

Relevant Coursework

- Big Data
- Data Mining
- Database Management
- Analysis of Algorithms
- Machine Learning
- Computer Vision
- Computer Network
- Natural Language Processing
- Data Science System
- High-Performance Distributed Computing

PROJECTS

Multi-Channel/Omni Channel Decision Engine:

Lead the design and implementation of a Multichannel/ Omnichannel decision engine that is connected into existing machine learning models to automate outputs from models, apply use case specific business rules and automate the upload to execution channels.

BYOML Platform for Personalized Cancer Treatment: Collaborated with oncologists to understand treatment protocols and patient data requirements and leveraged genomic data and patient histories to train models predicting treatment efficacy and Implemented secure model deployment pipelines on GCP, ensuring compliance with HIPAA regulations.

NBA Engine

Designed the solution which includes data pipeline to harmonize data from different sources available in Data Lake and integrate with Veeva/SFDC for sending suggestions to REP using REST API. Developed a inhouse product on Azure uses Machine Learning and generates next best actions for Medical Reps by analyzing the call attainment and sales anomalies based on historical data.

Health Care Incentive Compensation Application using IICS, Snowflake and Power BI : Led a team to implement an intelligent Incentive compensation application using IQVIA sales data utilized Databricks by applying ETL (Raw data) data then created multiple stored procedure for sales crediting , goal setting , incentive compensation planning and dashboarding in Power BI with row level restriction in one place with 100% accuracy and saving turnaround time by **630 %** by standardizing processes and achieving Skills: ETL Data Pipelines IICS Snowflake Python Power BI desktop and service.

Pharma Analytics Application using: Led a team to implement create a new system from scratch for reporting due to poor performance of legacy system. Implemented a robust ETL pipeline using Databricks, enabling seamless data integration into delta tables bronze table and applying cleaning to silver and business logic for gold table and applied machine earning for Segmentation of Grower, Loyalist, churner using their patient Journey

Patient at risk Discovery:

Developed a machine learning model to predict diabetes risk without requiring a doctor's visit, crucial during resource-strained periods like the COVID-19 pandemic. Employed algorithms such as Logistic Regression, k-NN, Decision Trees, Random Forests, and Ensemble Models to achieve high accuracy (95.5%) and sensitivity (97.6%). Key findings included identifying polydipsia and polyuria as strong predictors while obesity was not significant. The project aims to improve patient outcomes by making diabetes risk assessment more accessible and timelier.

Aspect extraction, Sentiment prediction, and recommend hotels using Spacy, Deep Learning, and NLP: Web scrap the reviews from online sources (trivago.com, booking.com) using Beautiful Soup. Analysis of the reviews and applied various techniques to identify the different aspects of the reviews and sentiment analysis of the reviews and recommendation engine using deep learning for a hotel recommendation.